CANDRON ENERGY INC.

FREE RESOURCE · PROJECT TOOLKIT

FAT & Commissioning Pack

Factory acceptance test agendas, pre-energization checklists and a punch list for medium-voltage switchgear and transformers.

A Factory Acceptance Test: MV Switchgear

B Factory Acceptance Test: Transformers

C Pre-Energization Checklist: MV Switchgear

D Pre-Energization Checklist: Transformers

E Punch List & Sign-Off

General guidance only. Follow the manufacturer's instructions, the project specification and the authority having jurisdiction. References: NETA ATS, CSA Z462, CSA C22.1, IEEE C37.20.2 and IEEE C57.12.90 / C57.12.91.

CANDRON Energy Inc. · Toronto, Ontario · candron.com · +1 (647) 616-2595

SHEET A · AT THE MANUFACTURER, BEFORE RELEASE FOR SHIPMENT

Factory Acceptance Test: MV Switchgear

A witness agenda for metal-clad and metal-enclosed switchgear. Agree the sequence with the manufacturer a week ahead so every test you want to see is scheduled while you are on site.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Before testing starts

- [ ] Approved drawings are the latest revision
  General arrangement, single-line, three-line, schematics and wiring diagrams.
- [ ] Test procedure and acceptance criteria shared in advance
- [ ] Calibration certificates for all test instruments are current
- [ ] Relay settings files received and match the approved coordination study
- [ ] Safety briefing held; test area barricaded; roles and sign-off authority agreed

Visual & mechanical inspection

- [ ] Nameplates and ratings match the approved drawings
  Voltage, continuous current, short-circuit rating, BIL, enclosure type.
- [ ] Enclosure, finish, doors, gaskets and hardware are complete and undamaged
- [ ] Compartment barriers and automatic shutters operate freely
- [ ] Bus joints are torque-marked; phase clearances look consistent
- [ ] Ground bus is continuous through every section
- [ ] Breakers rack smoothly to connected, test and disconnected positions
- [ ] Mechanical interlocks prevent racking a closed breaker and opening doors where specified
- [ ] Position indicators, operation counters and charging indicators work
- [ ] Space heaters and thermostats installed and wired
- [ ] Device labels and wire markers match the drawings

Electrical tests

- [ ] Insulation resistance of the main bus, phase-to-phase and phase-to-ground
- [ ] Power-frequency withstand (dielectric) test of the main circuit
  At the level required by the applicable standard; record voltage, duration and result.
- [ ] Contact resistance of breaker main contacts and bolted bus joints
- [ ] Current transformer ratio, polarity and secondary wiring checked
- [ ] Voltage transformer ratio, polarity and fusing checked
- [ ] Control wiring verified point to point against the schematics

Functional tests

- [ ] Breakers close and trip locally and remotely; anti-pump and trip-free work
- [ ] Each protective relay function picks up and trips the correct breaker
  Secondary injection against the settings file.
- [ ] Transfer and interlocking schemes run through every sequence
  Main-tie-main, automatic throw-over, generator transfer.
- [ ] Metering, alarms and annunciation read correctly
- [ ] SCADA and communications points verified end to end
- [ ] Control power transfer and battery or UPS supply tested

Close-out

- [ ] All deviations recorded on the punch list with an owner and a date
- [ ] Test reports signed by the manufacturer and the witness
- [ ] Release-for-shipment decision recorded
- [ ] Shipping splits, packing, lifting points and tie-downs reviewed

SHEET B · AT THE MANUFACTURER, BEFORE RELEASE FOR SHIPMENT

Factory Acceptance Test: Transformers

Routine tests for dry-type and liquid-filled transformers. Ask which tests will run while you are present and which will be reported; the losses and impedance results are the ones most often disputed later.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Before testing starts

- [ ] Approved drawings and the guaranteed losses, impedance and sound level on hand
- [ ] Test plan lists every routine test, plus any type or special tests ordered
- [ ] Calibration certificates for all test instruments are current

Visual inspection

- [ ] Nameplate matches the order
  kVA, voltages, BIL, impedance, connection and vector group, taps, cooling class.
- [ ] Bushings, terminations and cable or bus provisions are as drawn
- [ ] De-energized tap changer operates through every position and locks
- [ ] Accessories fitted: gauges, pressure relief, thermometers, fans, controls
- [ ] Core and tank or enclosure grounding provisions in place

Routine tests

- [ ] Winding resistance on every tap
- [ ] Turns ratio on every tap, with polarity and phase relation (vector group)
- [ ] Insulation resistance (and polarization index if specified)
- [ ] No-load loss and excitation current
- [ ] Load loss and impedance, corrected to the reference temperature
- [ ] Applied potential (separate-source) test
- [ ] Induced potential test
- [ ] Liquid-filled: leak test, plus fluid dielectric strength and moisture
- [ ] Liquid-filled: dissolved gas analysis baseline sample taken
- [ ] Cast-coil dry-type: partial discharge test, where specified

Accessories & controls

- [ ] Temperature alarms and trips set and tested
- [ ] Cooling fans start and stop on temperature and in manual
- [ ] Bushing CT ratio and polarity checked, where fitted
- [ ] Load tap changer and voltage regulator controls operate, where fitted

Close-out

- [ ] Results compared to guarantees and tolerances; any deviation recorded
- [ ] Test report signed by the manufacturer and the witness
- [ ] Shipping: impact recorder fitted, fluid or nitrogen status recorded, tie-downs reviewed

SHEET C · ON SITE, AFTER INSTALLATION AND BEFORE FIRST ENERGIZATION

Pre-Energization Checklist: MV Switchgear

Run this after the lineup is set, joined and cabled, and before the switching order is issued. It catches the problems that factory testing cannot: shipping damage, field joints, cables and settings.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Receipt & installation

- [ ] Shipping damage inspected; impact recorder reviewed and filed
- [ ] Lineup anchored and levelled per the installation drawings
- [ ] Shipping-split bus joints assembled and torqued; values recorded
- [ ] Ground bus connected to the station ground grid
- [ ] All covers and barriers reinstalled; no tools or debris inside

Cables & terminations

- [ ] Terminations made per the termination kit instructions
- [ ] Cable insulation tested as specified (insulation resistance or VLF)
- [ ] Phasing confirmed and phase identification applied
- [ ] Cable shields grounded as designed (one end or both ends)

Tests & settings

- [ ] Insulation resistance of the assembled bus
- [ ] Contact resistance of field bus joints
- [ ] Breakers operated; timing tested where specified
- [ ] Relay settings loaded match the approved settings file; trips verified by injection
- [ ] CT shorting blocks removed and every CT circuit closed through its burden
- [ ] VT fuses installed; trip circuits checked end to end
- [ ] Control power available; source transfer tested

Safety & approvals

- [ ] Arc-flash and equipment labels installed (CSA Z462)
- [ ] Lockouts cleared; energization plan and switching order approved
- [ ] Inspection authority and utility acceptance received
- [ ] As-left settings and test records filed with the owner

SHEET D · ON SITE, AFTER INSTALLATION AND BEFORE FIRST ENERGIZATION

Pre-Energization Checklist: Transformers

For padmount, substation and dry-type units. Most early-life transformer failures trace back to something on this list: moisture, a missed ground, the wrong tap or a loose termination.

PROJECT _____ EQUIPMENT / TAG _____ DATE _____ WITNESS _____

Receipt

- [ ] Impact recorder reading reviewed; no shipping damage or leaks
- [ ] Nitrogen pressure recorded, if shipped under gas
- [ ] Fluid level and temperature gauges read normally
- [ ] Core ground checked, where accessible

Installation

- [ ] Set level on the pad or base; clearances to walls and combustibles meet code
- [ ] Tank or enclosure and neutral grounded per the design
- [ ] Primary and secondary terminations torqued; values recorded
- [ ] Surge arresters installed where specified
- [ ] Dry-type: shipping blocks removed; coils and air ducts clean and clear

Tests

- [ ] Insulation resistance of the windings
- [ ] Turns ratio at the selected tap
- [ ] Winding resistance, if the unit was assembled or moved on site
- [ ] Liquid-filled: fluid sample after filling for dielectric strength and moisture

Settings

- [ ] De-energized tap set for the required secondary voltage and locked
- [ ] Temperature alarms and trips set and tested
- [ ] Fans tested in automatic and manual
- [ ] Pressure relief device and gauge drag pointers reset
- [ ] Load tap changer controls set, where fitted

First energization

- [ ] Secondary breakers open; energize at no load
- [ ] Observe at no load for the period in the energization plan
  Listen for unusual noise; watch temperatures and protection.
- [ ] Secondary voltages and phase rotation confirmed before load is applied
- [ ] First loading, temperatures and gauge readings recorded

SHEET E · FROM FAT THROUGH HANDOVER

Punch List & Sign-Off

One list for every open item, from the factory test to handover. Give every item a category, an owner and a date, and close it with a name.

A Must close before energization **B** Must close before handover **C** Minor or cosmetic

#	TAG	DESCRIPTION	CAT.	OWNER	DUE	CLOSED BY / DATE
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Manufacturer	Electrical contractor	Owner / engineer	Commissioning agent
NAME	NAME	NAME	NAME
SIGNATURE	SIGNATURE	SIGNATURE	SIGNATURE
DATE	DATE	DATE	DATE